

Introduction to Percolation Theory

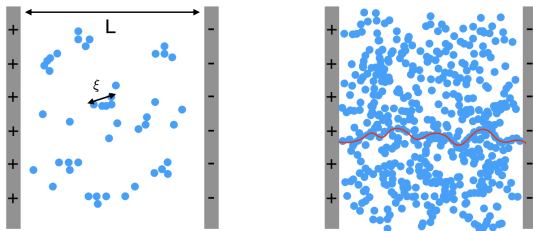
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October 2025

Basic physical phenomenon

EXAMPLE: On an insulating plate of size L , a density ρ of conducting metallic droplets is randomly deposited. At the border there is an electric potential difference. How does conductance depend on ρ ?



Increasing ρ , a non-zero conductance appears with a (phase) transition:

- 1 some physical quantity is strictly zero below a critical threshold and strictly non-zero above;
- 2 this is true only for $L \rightarrow \infty$.

Numerical experiments

Site percolation on a square lattice ($L = 100$)



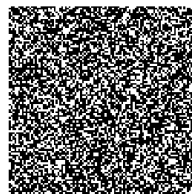
$p=0.1, s_{max} = 6$



$p=0.4, s_{max} = 47$



$p=0.55, s_{max} = 663$



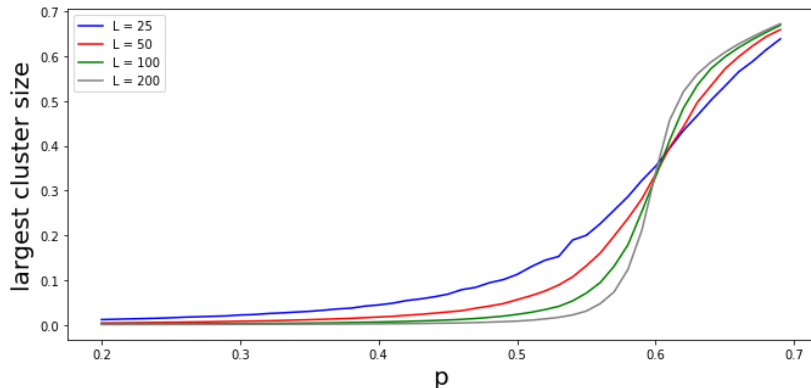
$p=0.62, s_{max} = 5049$

A (*connected*) **cluster** is a connected set of occupied sites:

increasing the occupation probability p , larger clusters appear.

Numerical experiments

Average fraction of sites in the largest cluster ($p_c \approx 0.59$)



Critical probability p_c : minimum value of occupation probability for which an infinite cluster exists in an infinite lattice.

Percolation in $d = 1$: trivial threshold

Independent site/bond occupation with probability p on a 1-dim lattice



The probability of a cluster of linear size L is

$$\Pi(p, L) = p^L \xrightarrow{L \rightarrow \infty} \begin{cases} 0 & p < 1 \\ 1 & p = 1 \end{cases} \quad (1)$$

A finite number of unoccupied sites (or closed bonds) is enough to prevent the existence of a percolating cluster, hence $p_c = 1$ in $d = 1$.

Percolation in $d = 1$: critical behavior

Probability that an arbitrary site (occupied or not) is, say, on the left border of a cluster of size ℓ is

$$n(\ell, p) = p^\ell (1 - p)^2. \quad (2)$$

For $p \rightarrow p_c$, writing $\Delta p = p_c - p = 1 - p$

$$n(\ell, p) = \Delta p^2 (1 - \Delta p)^\ell \approx \Delta p^2 e^{-\ell/\Sigma(p)} \quad (3)$$

with the *cluster size cut-off*

$$\Sigma(p) = -\frac{1}{\log(1 - \Delta p)} \approx (p_c - p)^{-\frac{1}{\sigma}}, \quad \sigma = 1 \quad (4)$$

Percolation in $d = 1$: critical behavior

Probability that an arbitrary site (occupied or not) belongs to a cluster:

$$\sum_{\ell \geq 1} \ell n(\ell, p) = \sum_{\ell \geq 1} \ell p^\ell (1-p)^2 \quad (5)$$

$$= (1-p)^2 p \frac{d}{dp} \sum_{\ell \geq 0} p^\ell \quad (6)$$

$$= (1-p)^2 p \frac{d}{dp} \frac{1}{1-p} \quad (7)$$

$$= p \quad (8)$$

Probability that an arbitrary occupied site i ($i \in \mathcal{O}$) belongs to any cluster of size ℓ ($i \in C_\ell$):

$$\mathbb{P}[i \in C_\ell | i \in \mathcal{O}] = \frac{\mathbb{P}[i \in C_\ell, i \in \mathcal{O}]}{\mathbb{P}[i \in \mathcal{O}]} = \frac{\ell n(\ell, p)}{p} \quad (9)$$

Percolation in $d = 1$: critical behavior

If we pick a random cluster, the selected site belongs to an ℓ -cluster with probability $\mathbb{P}[i \in C_\ell | i \in \mathcal{O}]$, so the *average cluster size* $S(p)$ is

$$S(p) = \sum_{\ell \geq 1} \ell \mathbb{P}[i \in C_\ell | i \in \mathcal{O}] \quad (10)$$

$$= \frac{(1-p)^2}{p} \sum_{\ell \geq 1} \ell^2 p^\ell \quad (11)$$

$$= \frac{(1-p)^2}{p} \sum_{\ell \geq 1} \left(p \frac{d}{dp}\right)^\ell p^\ell \quad (12)$$

$$= \frac{1+p}{1-p} \quad (13)$$

$$\implies S(p) \sim (p_c - p)^{-\gamma}, \quad \gamma = 1 \quad \text{for } p \rightarrow p_c^- \quad (14)$$

Percolation in $d = 1$: critical behavior

Probability that a site at distance r from an occupied site belongs to the same finite cluster (correlation function)

$$G(r) = p^r = e^{-r/\xi(p)} \quad (15)$$

where $\xi(p)$ is the *correlation length*,

$$\xi(p) = -\frac{1}{\log p} \quad (16)$$

$$\approx (p_c - p)^{-\nu}, \quad \nu = 1. \quad (17)$$

REMARK: In $d = 1$, $\Sigma(p) = \xi(p)$. In general, $\Sigma(p) \sim \xi(p)^{d_F}$, where $d_F \leq d$ is the fractal dimension of the percolation clusters.

Percolation in $d = 2$: exact results

A non-trivial bond percolation threshold exists in $d \geq 2$. For $d = 2$, simple bounds can be given:

$P(p)$ = probability that an arbitrary site O belongs to an infinite cluster.

PROP 1: If $p < \frac{1}{3}$, then $P(p) = 0$.

PROP 2: If $p > \frac{2}{3}$, then $P(p) > 0$.

Props 1 and 2 $\implies p_c < 1$ for bond percolation in $d = 2$. Similar results can be obtained for $d > 2$.

REMARK: The exact value for bond percolation on the two-dimensional square lattice is $p_c = 1/2$. Similar results exist for site percolation.

Percolation in $d = 2$: exact results

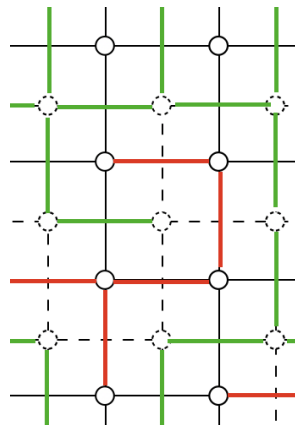
The exact threshold value can be obtained by a *duality argument*.

A dual bond is open (green) if the bond in the original lattice is closed (i.e. not red).

The occupation probability for dual bonds is $p_{\text{dual}} = q = 1 - p$.

By construction, the two objects do not intersect, so one or the other percolates.

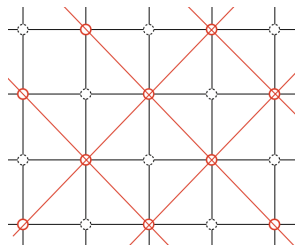
It follows that $q_c = p_c = 1/2$.



Percolation in $d = 2$: real-space Renormalization Group

If exact results are not accessible, one can also employ **real-space RG**:

- *coarse-graining step*: e.g remove alternate sites then consider “diagonal” bonds, create a bond in the new lattice if two sites were connected in the old one (i.e. by at least two open bonds joining them).
- *rescaling step*: since the number of sites $L^d \rightarrow L^d/2$, new lattice rescaled by a length factor $b = \sqrt{2}$ in $d = 2$.



REMARK: RG transformation is not exact, because bond probabilities are no longer independent after coarse-graining (higher-order interactions).

Percolation in $d = 2$: real-space Renormalization Group

The contributions to p' in the renormalized lattice are



$$p^4$$



$$4p^3(1-p)$$



$$2p^2(1-p)^2$$

which gives $p' = 2p^2 - p^4 \equiv \mathcal{R}_b(p)$.

Solving $p = \mathcal{R}_b(p)$, one gets the *RG fixed points*:

$$p^* = 0, \quad p^* = 1, \quad p_c = (\sqrt{5} - 1)/2 \approx 0.62. \quad (18)$$

The **RG flow diagram** follows from their stability properties



Percolation in $d = 2$: real-space Renormalization Group

Calculation of the *correlation length exponent* ν (exact result, $\nu = 4/3$)

After RG transformation, $\xi' = \frac{\xi}{b}$,

$$|\mathcal{R}_b(p) - p_c|^{-\nu} = \frac{|p - p_c|^{-\nu}}{b} \quad \Rightarrow \quad \nu = \frac{\log b}{\log \left[\frac{|\mathcal{R}_b(p) - p_c|}{|p - p_c|} \right]} \quad (19)$$

Since $\mathcal{R}_b(p_c) = p_c$, for $p \rightarrow p_c$ one can approximate

$$\frac{|\mathcal{R}_b(p) - \mathcal{R}_b(p_c)|}{|p - p_c|} \approx \left| \frac{d\mathcal{R}_b(p)}{dp} \right|_{p=p_c} \quad (20)$$

$$= 4p_c (1 - p_c^2) = (\sqrt{5} - 1)^2 \quad (21)$$

$$\Rightarrow \quad \nu = \frac{\log 2}{4 \log (\sqrt{5} - 1)} \approx 0.81 \quad (22)$$

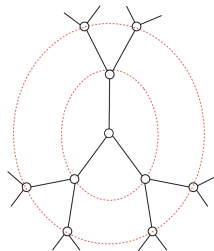
Not so good, but the RG scheme can be improved.

Bethe lattice ($d = \infty$)

Bethe lattice $\mathcal{B}$ is an infinite regular tree, recursively defined.

The finite Bethe lattice $\mathcal{B}_n$ (or Cayley tree) of coordination number z is generated as follows:

- 1 start with a root (shell $\mathcal{S}_0$)
- 2 add z new neighboring sites to the root (shell $\mathcal{S}_1$)
- 3 for each site in shell $\mathcal{S}_\ell$ for $1 \leq \ell \leq n-1$ add $z-1$ new neighboring sites (forming shell $\ell+1$).



REMARK: It is an example of regular structure in infinite dimensions.

Bethe lattice ($d = \infty$)

The Bethe lattice has properties of infinite dimensional regular lattices:

- the border-to-bulk mass ratio is finite

$$|\mathcal{S}_\ell| = z(z-1)^{\ell-1} \quad (23)$$

$$|\mathcal{B}_n| = 1 + \sum_{\ell \geq 1}^n |\mathcal{S}_\ell| = \frac{z(z-1)^n - 2}{z-2} \quad (24)$$

$$\implies \frac{|\mathcal{S}_n|}{|\mathcal{B}_n|} = \frac{z(z-2)(z-1)^{n-1}}{z(z-1)^n - 2} \xrightarrow{n} \frac{z-2}{z-1} > 0 \quad (25)$$

In any Euclidean space,

$$\frac{|\mathcal{S}_n|}{|\mathcal{B}_n|} \propto \frac{L^{d-1}}{L^d} = V^{-1/d} \xrightarrow{V \rightarrow \infty} 0 \quad (26)$$

except for strictly $d = \infty$.

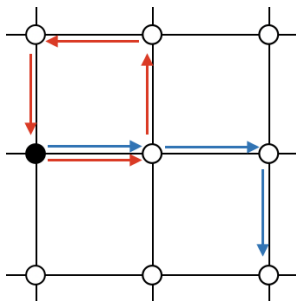
Bethe lattice ($d = \infty$)

The Bethe lattice has properties of infinite dimensional regular lattices:

- no correlation induced by loops

Consider paths of length $\ell = 4$ starting from an arbitrary node in d dimensions,

- the number of paths scales as $2d(2d - 1)^2$,
- number of loops is at most $2d(2d - 2)$.



The fraction of loops of length ℓ vanishes as $d \rightarrow \infty$. True $\forall \ell$

Percolation on Bethe lattice: critical behavior

Sites (or equivalently bonds) are independently occupied with probability p .

Percolation threshold located exactly exploiting recursive tree structure.

Probability $Q(p)$ that a randomly chosen edge does not lead to an infinite cluster

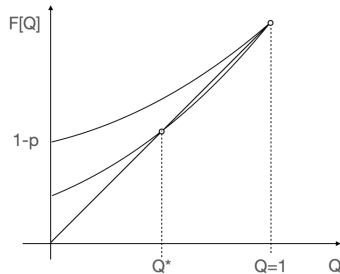
$$Q(p) = 1 - p + pQ(p)^{z-1} \equiv F[Q] \quad (27)$$

Solutions: $Q(p) = 1$ always exists, but
 $\exists Q^* < 1$ if

$$F'[Q]|_{Q=1} = (z-1)p \quad (28)$$

$$> 1 \quad (29)$$

which implies $p_c = \frac{1}{z-1}$.



Percolation on Bethe lattice: critical behavior

For $p \rightarrow p_c^+$, assume $Q(p) = 1 - \epsilon$

$$1 - \epsilon = 1 - p + p[1 - \epsilon]^{z-1} \quad (30)$$

$$= 1 - p + p \left[1 - (z-1)\epsilon + \frac{(z-2)(z-1)}{2} \epsilon^2 \right] \quad (31)$$

$$\implies \epsilon = \frac{2}{p(z-2)} (p - p_c) \approx \frac{2(z-1)}{z-2} (p - p_c) \quad (32)$$

Probability $P(p)$ that a *node belongs to the infinite (percolating) cluster*

$$P(p) = p[1 - Q(p)^z] \quad (33)$$

$$= p[1 - (1 - \epsilon)^z] \simeq pz\epsilon \approx \frac{2z}{z-2} (p - p_c) \quad (34)$$

$$\implies P(p) \sim (p - p_c)^\beta, \quad \beta = 1 \quad (35)$$

Percolation on Bethe lattice: critical behavior

The average size $T(p)$ of the finite cluster an arbitrary edge belongs to satisfies (for $p < p_c$)

$$T(p) = p [1 + (z - 1) T(p)] \quad \implies \quad T(p) = \frac{p}{1 - (z - 1)p} \quad (36)$$

Average cluster size $S(p)$:

$$S(p) = [1 + zT(p)] p \quad (37)$$

$$= \left[\frac{1 - p(z - 1) + zp}{1 - p(z - 1)} \right] p \quad (38)$$

$$= \frac{pp_c(1 + p)}{p_c - p} \quad (39)$$

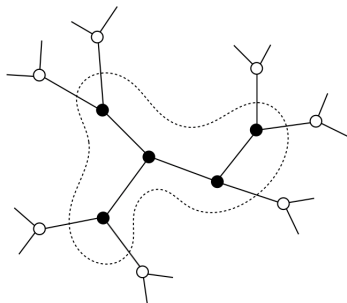
$$\propto (p_c - p)^{-\gamma}, \quad \gamma = 1 \quad (40)$$

Percolation on Bethe lattice: cluster size distribution

The occupied cluster is a tree of size s :

- $s - 1$ internal edges
- num. external edges

$$\begin{aligned}t &= z + (s - 1)(z - 1) - (s - 1) \\ &= s(z - 2) + 2\end{aligned}$$



Probability that a node is the root of a cluster of size s

$$n(s, p) = g_s p^s (1 - p)^t = g_s p^s (1 - p)^{s(z-2)+2} \quad (41)$$

but the coefficient g_s is difficult to compute!

Percolation on Bethe lattice: cluster size distribution

To avoid computing g_s , one can consider the ratio

$$\frac{n(s, p)}{n(s, p_c)} = \left(\frac{1-p}{1-p_c} \right)^2 \left[\frac{p(1-p)^{z-2}}{p_c(1-p_c)^{z-2}} \right]^s \quad (42)$$

$$= \left(\frac{1-p}{1-p_c} \right)^2 e^{-s/\Sigma(p)} \quad (43)$$

with

$$\Sigma(p) = \left[\log \left(\frac{p_c(1-p_c)^{z-2}}{p(1-p)^{z-2}} \right) \right]^{-1} \quad (44)$$

It can be shown that

$$\Sigma(p) \propto |p - p_c|^{-1/\sigma}, \quad \sigma = \frac{1}{2} \quad (45)$$

Percolation on Bethe lattice: cluster size distribution

To compute $\Sigma(p)$, focus on the argument of the log

$$\frac{p}{p_c} \left(\frac{1-p}{1-p_c} \right)^{z-2} = \frac{p_c + \Delta p}{p_c} \left(\frac{1-p_c - \Delta p}{1-p_c} \right)^{z-2} \quad (46)$$

$$= \left(1 + \frac{\Delta p}{p_c} \right) \left(1 - \frac{\Delta p}{1-p_c} \right)^{z-2} \quad (47)$$

$$\approx \left(1 + \frac{\Delta p}{p_c} \right) \left[1 - \frac{z-2}{1-p_c} \Delta p + \frac{(z-3)(z-2)}{2(1-p_c)^2} (\Delta p)^2 \right] \\ = 1 - A(z) (\Delta p)^2 \quad (48)$$

with $A(z) = \frac{(z-1)^3}{2(z-2)}$. Hence,

$$\Sigma(p) = \left[\log \frac{1}{1 - A(z) (\Delta p)^2} \right]^{-1} = \left[A(z) (\Delta p)^2 \right]^{-1} \propto |p - p_c|^{-2} \quad (49)$$

Percolation on Bethe lattice: cluster size distribution

Let's compute the power-law behavior $n(s, p) \sim s^{-\tau}$ for $p \simeq p_c$.

Since $\sum_{s=1}^{\infty} s^k \rightarrow \infty$ for $k \geq -1$:

- the condition

$$\sum_{s=1}^{\infty} s n(s, p_c) = p_c < +\infty, \quad (50)$$

implies $1 - \tau < -1$, or $\tau > 2$;

- the condition

$$S(p) = \frac{\sum_{s=1}^{\infty} s^2 n(s, p)}{\sum_{s=1}^{\infty} s n(s, p)} \xrightarrow{p \rightarrow p_c^-} +\infty, \quad (51)$$

implies $2 - \tau \geq -1$, or $\tau \leq 3$.

In summary,

$$2 < \tau \leq 3. \quad (52)$$

Scaling laws and relations

The exponent τ can be found using the scaling properties of $S(p)$

$$S(p) \propto \sum_{s=1}^{\infty} s^{2-\tau} e^{-s/\Sigma(p)} \quad (53)$$

$$\approx \int_1^{+\infty} ds s^{2-\tau} e^{-s/\Sigma(p)} \quad (54)$$

$$= \Sigma(p)^{3-\tau} \int_{1/\Sigma(p)}^{+\infty} dx x^{2-\tau} e^{-x} \quad (55)$$

$$= \Sigma(p)^{3-\tau} \Gamma(3-\tau) \quad (56)$$

$$\propto |p_c - p|^{-\frac{3-\tau}{\sigma}} \quad (57)$$

Since $S(p) \sim |p_c - p|^{-\gamma}$, we found the **scaling relation**

$$\gamma = \frac{3-\tau}{\sigma} \quad (58)$$

Scaling laws and relations

For $d = 1$, it gives $\gamma = 1$, $\sigma = 1$ which imply $\tau = 2$ but the exact expression is

$$n(s, p) = (p_c - p)^2 e^{-s/\Sigma(p)}, \quad (59)$$

with no power law in s . Is it $\tau = 0$ or $\tau = 2$?

The distribution can be written as follows

$$n(s, p) = (p_c - p)^2 e^{-s/\Sigma(p)} \quad (60)$$

$$= s^{-2} [(p_c - p) s]^2 e^{-s(p_c - p)} \quad (61)$$

$$= s^{-\tau} F[(p_c - p) s^\sigma], \quad (62)$$

with $\tau = 2$ and $\sigma = 1$, and $F(\cdot)$ is a scaling function.

Scaling laws and relations

The **scaling function** $F(z)$ is such that

$$F(z) = \begin{cases} \begin{cases} z^2 & d = 1 \\ \text{const.} & d > 1 \end{cases} & |z| \ll 1 \\ \text{decays exp.} & |z| \gg 1 \end{cases} \quad (63)$$

where $|z| \ll 1$ (resp. $|z| \gg 1$) means $s \ll \Sigma(p)$ (resp. $s \gg \Sigma(p)$).

The scaling behavior can be obtained using dimensional analysis:
since $[n(s, p)] = [s^{-\tau}]$ then

$$n(s, p) = s^{-\tau} \tilde{f}(s, p, p_c), \quad (64)$$

and $[s] = (p - p_c)^{-1/\sigma}$ implies

$$n(s, p) = s^{-\tau} F[(p - p_c)s^\sigma]. \quad (65)$$

Scaling laws and relations

Another scaling relation, involving exponent β , can be obtained from the scaling behavior of $P(p)$ in the limit $p \rightarrow p_c^+$,

$$P(p) = p - \sum_{s=1}^{\infty} s n(s, p) \quad (66)$$

$$= p - p_c - \sum_{s=1}^{\infty} s (n(s, p) - n(s, p_c)) \quad (67)$$

$$= \Delta p + \sum_{s=1}^{\infty} s^{1-\tau} \left(1 - e^{-s/\Sigma(p)}\right) \quad (68)$$

$$= \Delta p + \Sigma(p)^{2-\tau} \int_{1/\Sigma(p)}^{\infty} dx x^{1-\tau} (1 - e^{-x}) \quad (69)$$

$$= \Delta p + \Sigma(p)^{2-\tau} \int_0^{\infty} dx x^{1-\tau} (1 - e^{-x}) \quad (70)$$

Scaling laws and relations

Integrating by parts

$$P(p) = \Delta p + \Sigma(p)^{2-\tau} \left\{ \left[\frac{x^{2-\tau}}{2-\tau} (1 - e^{-x}) \right]_0^\infty - \int_0^\infty dx \frac{x^{2-\tau}}{2-\tau} e^{-x} \right\} \quad (71)$$

$$= \Delta p + \Sigma(p)^{2-\tau} \left[0 + \frac{1}{\tau-2} \Gamma(3-\tau) \right] \quad (72)$$

$$= \Delta p + \frac{\Gamma(3-\tau)}{\tau-2} \Sigma(p)^{2-\tau} \quad (73)$$

Using $\Sigma(p) \sim (p - p_c)^{-1/\sigma}$ and $P(p) \sim (p - p_c)^\beta$,

$$\implies \beta = \frac{\tau-2}{\sigma}, \quad p \rightarrow p_c^+. \quad (74)$$

Structure of percolation clusters: fractal nature

Correlation function on a Bethe lattice

$$G(r) = z(z-1)^{r-1} p^{r+1} \propto e^{-r \log p_c/p} \quad (75)$$

then

$$\xi = \frac{1}{\log \frac{p_c}{p}} \approx (p_c - p)^{-\nu_{\text{Bethe}}}, \quad \nu_{\text{Bethe}} = 1 \quad p \rightarrow p_c \quad (76)$$

REMARK: The divergence of the correlation length is valid from above as well, if we focus on finite clusters.

Is $\nu_{\text{Bethe}} = 1$ the correct *mean-field value*? NO, it depends on measuring correlation using chemical distance instead of Euclidean distance.

For $d \rightarrow \infty$, percolation paths are random walks, i.e. $\xi^2 \sim (p_c - p)^{-\nu_{\text{Bethe}}}$

$$\implies \nu_{MF} = \nu_{\text{Bethe}}/2 = \frac{1}{2} \quad (77)$$

Structure of percolation clusters: fractal nature

Suppose the incipient percolating cluster is a fractal close to p_c , then

$$\Sigma(p) \sim \xi^{d_F}. \quad (78)$$

To determine d_F , use $\xi \sim (p_c - p)^{-\nu}$ and $\Sigma(p) \sim (p_c - p)^{-1/\sigma}$, then

$$d_F = \frac{1}{\nu\sigma} \quad (79)$$

and using the MF results ($\nu = 1/2$, $\sigma = 1/2$) we get $d_F = 4$.

REMARK: Suppose $p > p_c$, the percolating cluster is a fractal with fractal dimension $d_F < d$ only for scales $\ell \ll \xi$, instead at scales $\ell \gg \xi$, it is not a fractal ($d_F = d$).

Structure of percolation clusters: hyperscaling

How does the spatial dimension enters scaling relations?

Suppose we take a box of linear size L , for $p > p_c$:

- if $L \ll \xi$, then $\Sigma(p) \propto L^{d_F}$ with a prefactor independent of $p - p_c$,
- if $L \gg \xi$, then $\Sigma(p) \propto P(p)L^d$,

For $L \approx \xi$, the condition $L^{d_F} \sim L^d P(p)$ gives

$$\xi^{d_F} \sim P(p)\xi^d \implies (p - p_c)^{\beta - \nu d} \sim (p - p_c)^{-\nu d_F} \quad (80)$$

that is the ***hyperscaling relation***

$$d_F = d - \beta/\nu. \quad (81)$$

REMARK: Using $d_F = \frac{1}{\nu\sigma}$, $1/\sigma = \gamma/(3 - \tau) = \beta/(\tau - 2)$, the hyperscaling relation can be written as

$$d\nu = \gamma + 2\beta \quad (82)$$

Critical behavior: summary

Replacing the MF exponents, the hyperscaling relation identifies the **upper critical dimension** $d_{uc} = 6$.

For $d \geq d_{uc}$, the *mean-field critical behavior* is expected.

d	1	2	3	...	≥ 6
β	-	$\frac{5}{36}$	0.4273		1
γ	1	$\frac{43}{18}$	1.8357		1
ν	1	$\frac{4}{3}$	0.8774		$\frac{1}{2}$
σ	1	$\frac{36}{91}$	0.4419		$\frac{1}{2}$
τ	2	$\frac{187}{91}$	2.1892		$\frac{5}{2}$

Why $d_{uc} = 6$? Loops are created by pairs of sites of the percolating cluster connected by a random path:

$$n_{\text{loops}} \sim \xi^d \left(\xi^{d_F - d} \right) \left(\xi^{2-d} \right) \rightarrow 0, \quad d \geq 6 \quad (83)$$